
The First Lever, and an Audit That Withdrew Its Explanation

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Abstract

After sixteen closed directions we report the first intervention to exceed this frame’s adoption gate: reducing model depth from 8 to 6 (with width following the aspect ratio, $768 \rightarrow 640$) gives a paired -0.003252 val_bpb at $n=4$, 4/4 seeds, $t = -4.70$, against a gate of 0.002614. We do not claim adoption here: the frame requires $n=10$, and the campaign’s own E2 precedent read -0.000772 with $t = -1.96$ at $n=4$ before collapsing to null at $n=9$. We also report, and this is the paper’s more useful half, an audit of the token law $\text{val_bpb} = a - b \ln(\text{tokens})$ that the campaign has used for eight blocks as a break-even calculator, a control variate, and a decomposition tool. Fitted on $n=50$ identical-configuration control arms it returns $b = 0.0710 \pm 0.0034$ with $R^2 = 0.972$ — but the clean arms all cluster at 2000–2030 steps, so essentially all slope information comes from two contention-damaged arms; deleting one point takes R^2 to 0.676, and restricting to the clean cluster gives $R^2 = 0.380$. Its validity window is 1131–2032 steps, and outside it the relation does not lose accuracy but inverts: at the same configuration a $2\times$ -budget run completes 3918 steps and scores 0.9870 where the fit predicts 0.8854, an error of +0.102 (39 gates). Depth 6 runs at 3279 steps, $1.61\times$ beyond the fitted range. We therefore withdraw our own decomposition of the depth-6 result — a -0.0292 token credit against a $+0.0259$ intrinsic cost — as unsupported extrapolation, while the raw paired measurement, which invokes no law, stands.

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1. Why Depth Was the Sixteen-Failure Prediction

Papers 026–028 measured two magnitudes in this frame: intrinsic quality effects $\approx 1e-3$, throughput effects $\approx 1e-2$. That pair is not merely discouraging, it is a selection rule. An intervention can clear a $2.6e-3$ gate only if it moves a $1e-2$ -scale term, and every lever tested moved at most one such term — refinements moved only the intrinsic term, kernel work moved only the throughput term and was mostly already captured by torch.compile.

Model shape moves both at $1e-2$. It was the only untested class with that property, and the dense parameters are just $\approx 5\%$ of this model — the hashed n -gram tables hold $\approx 1.61e9$ and are untouched by depth — so the capacity surrendered is a small slice while the throughput bought is large.

2. Result

Depth $\in \{5, 6, 7, 8\}$, width following ASPECT_RATIO, 4 seeds, fixed 300s, 16/16 arms clean on the duty-cycle screen.

depth	dim	val_bpb	steps	vs d8 (paired)
8	768	0.932703	2016	—
7	768	0.931571	2245	-0.001132
6	640	0.929451	3279	-0.003252
5	512	0.978424	4528	$+0.045880$

Depth 6: 1.24 gates, 4/4 seeds, $t = -4.70$. Not adopted; E11 extends to $n=10$.

The depth floor is the data wall, not capacity. We initially read depth 5’s collapse as lost capacity. The logged epoch field says otherwise: depth 5 is the only arm reaching a third pass over the frozen corpus, and its 0.9784 sits beside the recorded epoch-3 collapse (exp347, 0.987). Depth 6 sits just under the wall at 484M tokens. We noticed only because epoch was pulled alongside val_bpb rather than after it.

3. The Audit, and What It Cost Us

The token law has been load-bearing for eight blocks. Prompted by the depth-6 numbers we audited it against $n=50$ control arms — identical configuration, identical data, differing only in how many steps contention allowed, so the intrinsic term is exactly zero by construction.

data	b	R^2
all 50	0.0710	0.972
drop the 1131-step point	0.0745	0.676
clean cluster only (48)	0.0993	0.380

The R^2 is carried by one point. The clean arms occupy a single x -value; this is two x -values with 48 replicates, not a designed sweep.

The validity window is hard-edged. Fitted range 1131–2032 steps. At the same depth-8 configuration, exp347_sota2x runs 3918 steps and scores 0.9870 against a predicted 0.8854: error +0.102, 39 gates, and the sign of the token effect has flipped. Past two passes, more tokens make the model worse.

The intercept does not transport. Predicting depth 6 from the depth-8 fit gives 0.8976 against an observed 0.9295 — a +0.0318 (12-gate) shift, which is precisely the intrinsic term the law cannot see.

What we withdraw. One block ago we reported depth 6 as a -0.0292 token credit against a $+0.0259$ intrinsic cost, and concluded “the shallower model is ≈ 10 gates worse per token and wins only because throughput overpays.” Depth 6 runs at 3279 steps, $1.61\times$ past the fitted range and inside the region where the same-configuration $2\times$ run shows the relation already inverted. That decomposition is withdrawn. It also resolves a discrepancy we flagged but could not explain: the -0.029 credit versus a recorded -0.022 ceiling on all throughput work. Neither number was wrong about the world; the credit was an invalid quantity.

What survives. $\text{val_bpb}(d6) - \text{val_bpb}(d8) = -0.003252$, $4/4$, $t = -4.70$: a direct equal-wall-clock paired A/B invoking no law. E11 measures the same raw quantity at $n=10$. The belief record is downgraded to challenged with the window, the leverage structure, and the valid/invalid uses written into it.

4. Next Experiments

Per docs/EXPERIMENT_WORKFLOW.md, each states Claim, Mechanism, Hypothesis, Reasoning, then ratings (1–5 on novelty / provenance / validity / impact / reliability / feasibility / falsifiability).

4.1. E12 — Designed token sweep at fixed configuration

- Claim. blf_token_law_fixed_batch_provisional, now challenged. Internal, self-proposed; no external claim is invoked. Its recorded evidence strength is weak and its coefficient was fit post hoc.
- Mechanism. Observed phenomenon: val_bpb tracks $\ln(\text{steps})$ tightly among damaged arms, yet a same-configuration $2\times$ -budget run lands $+0.102$ above the extrapolation. Proposed cause: two regimes with opposite signs superposed — a data-acquisition regime where fresh tokens reduce loss logarithmically, and a repetition regime where re-reading a frozen corpus increases val loss. The observed curve is their sum, so it is monotone decreasing until re-reads dominate and then turns up. Rival account: a single smooth log law plus contention noise, with exp347 explained by something incidental to the $2\times$ budget. Separating observable: the shape between 2000 and 3900 steps — superposition predicts a minimum inside that interval, the single-law account predicts monotone decrease throughout.
- Hypothesis. Vary only TIME_BUDGET at fixed depth 8 and fixed batch over 5 points spanning ≈ 1000 –3800 steps, 3 seeds each. Estimand: val_bpb as a function of $\ln(\text{steps})$ within one configuration. Prediction: a minimum at 2600 ± 500 steps, and b estimated on the pre-minimum points within $[0.046, 0.086]$. Decision rule: if val_bpb is monotone decreasing to 3800 steps, the superposition mechanism is refuted and exp347 must be re-examined for a confound. Trap-check: none needed; every point is informative. Analysis rule: no control variate — this experiment is the control variate’s calibration.
- Reasoning. Every throughput decision for eight blocks has used a slope whose R^2 rests on one contention-damaged arm, and the depth-6 interpretation already broke on it. Evidence for: three same-configuration $2\times$ runs agree closely (0.9870, 0.9860, and 0.980 at a slightly different config). Evidence against: the minimum may sit outside the reachable range, leaving the boundary bracketed rather than located — which would still be more than we have now. This is infrastructure, not a win, and should be judged as such.
- Ratings. 1 / 5 / 5 / 4 / 5 / 5 / 5.

4.2. E13 — Quality-per-token at depth 6

- Claim. Internal and self-proposed, derived from E10’s own epoch field; no external claim.
- Mechanism. Observed phenomenon: depth 6 reaches 484M tokens, just under the third-pass boundary; depth 5 crosses it and loses 17.6 gates. Proposed cause: at depth 6 the binding constraint has switched from compute to data, so marginal tokens are re-reads and carry little new signal. Consequence: the throughput channel, which paid for every gain up to here, is exhausted at this operating point. Rival account: depth 6 is still compute-bound and more speed would still pay. Separating observable: apply any throughput-only lever at depth 6 — superposition predicts ≈ 0 or negative, compute-bound predicts the usual token-law credit.
- Hypothesis. At depth 6, run a throughput-only change against a depth-6 control, $n=10$ paired. Prediction, registered: the gain is ≤ 0 , in contrast to the same change at depth 8. Decision rule: a gain ≥ 1 gate at depth 6 refutes the data-wall reading of E10 and reopens throughput. Analysis rule: RAW is the decision metric (the treatment’s mechanism is throughput); the control variate must not be applied.
- Reasoning. This is the falsifier for the story we are about to tell about depth 6, and it is cheap. Evidence for: depth 5’s collapse coincides exactly with the epoch-3 crossing rather than with any capacity threshold. Evidence against: the epoch counter is coarse (integer passes), so “just under the wall” is a claim about a boundary we have located only to within one pass — E12 sharpens exactly this.
- Ratings. 3 / 4 / 4 / 4 / 4 / 5 / 5.

4.3. E14 — Depth 6 with the freed memory spent on batch

- Claim. Internal; adjacent to the recorded Block-17 factorial finding that batch/step-count dominates prior levers. Scope match adequate (same frame), evidence class internal-quarantined.
- Mechanism. Observed phenomenon: depth 6 uses 44 GB against depth 8’s 52.6 GB, and its step is 91.3 ms against 149.6 ms. Proposed cause: shape reduction frees both memory and time, and in a data-limited regime the useful way to spend freed memory is a larger batch, which raises tokens per step without raising re-reads per token — unlike more steps, which is what E13 predicts is now

worthless. Rival account: batch and depth interact only through total tokens, in which case a larger batch is just another throughput lever and E13’s prediction applies to it too.

- Hypothesis. Depth 6 at `DEVICE_BATCH_SIZE` $\in \{72, 96\}$, $n=10$ paired against depth-6 baseline. Prediction: if batch is a distinct axis, ≥ 1 gate; if it is only throughput, ≤ 0 per E13. Decision rule: standard funnel at the 0.002614 gate. Trap-check: confirm the larger batch fits in memory and does not reduce step count more than proportionally.
- Reasoning. Evidence for: the recorded token law is explicitly not sufficient across batch sizes — the paired $B=96$ rows falsified that extrapolation (mean -0.000406 , $2/5$ same sign), which is precisely the signature of batch being a separate axis rather than a token proxy. Evidence against: that same result showed no benefit at depth 8, so the proposal rests entirely on the interaction with the freed memory at depth 6, which is untested.
- Ratings. 3 / 3 / 4 / 4 / 3 / 4 / 5.

4.4. E15 — Extend the hash-prime families and test depth 9–12

- Claim. Internal; no external claim. Motivated by a known implementation limit rather than by a hypothesis about the world.
- Mechanism. Observed phenomenon: the depth sweep is one-sided — depths above 8 raise `IndexError` because the per-layer decorrelated hash-prime families are hardcoded lists of 4 (bigram) and 3 (trigram) tuples. Consequence: we have bracketed the optimum from below (depth 5 collapses) but not from above, so “depth 6 is optimal” is an inference from three points on one side.
- Hypothesis. Extend the prime families, then run depths $\{9, 10, 12\}$ at 4 seeds. Prediction: all are worse than depth 8, since they are slower and the token law’s decreasing branch still holds below 2000 steps. Decision rule: if any deeper model beats depth 8, the shape story is not monotone and the whole sweep needs re-reading.
- Reasoning. Evidence for: cheap, and it removes a limitation we have had to state in every record since E10. Evidence against: low expected value — deeper means fewer steps, and the token law’s decreasing branch is the one regime we are confident about. Listed for completeness of the optimum claim, not because we expect a win. Adding prime families also changes the hashing structure

for existing depths, so it must be verified byte-identical at depth 8 before use.

- Ratings. 1 / 5 / 4 / 2 / 4 / 3 / 5.

5. Conclusion

Sixteen failures produced a selection rule, the selection rule produced the campaign's first gate-clearing candidate, and auditing that candidate's explanation cost us the explanation. Depth 6 is real at $n=4$ and pending at $n=10$; the account of why it works is now open, because the tool we used to build that account turns out to be valid only inside a window the result sits outside of. We regard reporting the surviving measurement and the withdrawn interpretation separately as the substance of the result, not an aside to it.